ENERTECH MASTER KNOWLEDGE DOCUMENT (v2 — Expanded with Website Data)

Sources: EnerTech internal knowledge PDF + enertechups.com + enercube.in (fetched 25 Jul 2026). Where the original PDF had `[DATA_PENDING]`, values below are now filled from the official website. Where the two source websites disagree on a fact (noted inline), both values are shown so your team can confirm the correct one before publishing.

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1. COMPANY CONTEXT LAYER

1.1 EnerTech UPS Pvt. Ltd.

- EnerTech UPS Pvt. Ltd. is a solar and power-electronics designer and manufacturer headquartered at SR. No. 399/1-2, Plot No. 5, Bhare, P.O. Ghotawade, Near Hinjawadi IT Park, Pune – 412115, Maharashtra, India.
- Founding year: the company's leadership page (CEO Kaushik Deshpande) states EnerTech was **established in 1989** and is referred to as "35+ years" of experience across most of the site. One "About Us" panel on the same page instead states **1999**. Treat **1989** as the primary figure (it is the one repeated across product pages and matches "35+ years"); flag the 1999 mention to the client for correction.
- R&D: the company operates a **DSIR-recognised R&D laboratory**, certified by the Government of India.
- Scale: **35,000+ installations** worldwide, **600+ MW** of total installed capacity (per the homepage regional breakdown: India ~480 MW, Middle East ~60 MW, Africa ~30 MW, Southeast Asia ~20 MW, Others ~10 MW — note the earlier PDF cited "150 MW globally," which the current website supersedes with 600+ MW).
- Certifications referenced across the site: **BIS approved**, **IEC certified**, **ISO 9001:2015 certified**, CE certified (on BESS/PCS products), IEEE 1547, IEC 62477, IEC 61000, UL 1741 (bidirectional PCS line specifically).
- Leadership: Vijay Deshpande — Chairman; Kaushik Deshpande — CEO.
- Core domains: Battery Energy Storage Systems (BESS), industrial/critical-power UPS, solar hybrid PCUs, static frequency converters, industrial battery chargers, servo voltage stabilizers.
- Typical served sectors: residential, commercial, industrial (C&I), telecom microgrids, oil dispensing/petrol stations, cold storage, healthcare, data centers, hospitality, textiles, pharmaceuticals, and agriculture/rural electrification.

1.2 EnerCube (enercube.in) — Group BESS Brand

- EnerCube is EnerTech's dedicated **Battery Energy Storage System (BESS)** brand/entity, describing itself as a high-tech enterprise for energy-conversion products (BESS, Energy Storage PCS, Hybrid Solar PCU, and other power electronics).

- Site-reported scale: **31+ years' experience**, **30,000+ happy customers**, **120+ MW installed**, **40+ channel partners**.
- Registered office: S. No. 399/1-2, Bhare, P.O. Ghotawade, Near Pirangut, Taluka Mulshi, Dist. Pune – 412115 (same industrial address as EnerTech UPS).
- Leadership shown on enercube.in: Anurag Mathur — Founder & Director; Kaushik Deshpande — Director; Vijay Deshpande — Founder & Chairman (same chairman as EnerTech UPS, confirming EnerCube is part of the EnerTech group).
- Reference clients named on the EnerCube site: Adani, BHEL, TATA, TAFE, HPCL, NPCIL.
- EnerCube contact: +91 9920051821 / +91 7387422231, sales@enercube.in.
- EnerCube product lines: Mini e-Storage, Plug & Play Storage Cabinet, All-in-One Hybrid e-Storage, MWh Energy Storage Container, Hybrid Eco Series BESS — these map to the same BESS products listed under enertechups.com's "Solar BESS" menu.

1.3 Official Contact Directory (enertechups.com)

Department	Phone	Email
Commercial / Sales Enquiry	+91 9373336340 / +91 9822407189	sales@enertechups.com
Customer Care & Support (service)	+91 9373679255	support@enertechups.com
Industrial / Bulk Enquiry	+91 9373679255 / +91 9075088292	industrial@enertechups.com
Careers / HR	+91 9075088292	recruitment@enertechups.com / hr@enertechups.com
EnerCube (BESS brand)	+91 9920051821 / +91 7387422231	sales@enercube.in

Note: your working prompt currently sends sales leads to "Mr. Shivaji Chouhan on 9370659050" and service leads to WhatsApp 9373679255 / support@enertechups.com. The 9373679255 number matches the official "Customer Care & Support" line on the live site, so that part is confirmed correct. 9370659050 does not appear anywhere on the current site — it may be a personal/regional number for Mr. Chouhan that isn't published, which is fine, but worth double-checking it's still active.

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2. FULL PRODUCT RANGE (as currently listed on enertechups.com)

Product Category	Power Range (site-confirmed)	Notes
Solar Hybrid Inverter (1Ph) — SunMagic	5 kVA – 40 kVA	1-Phase
Solar Hybrid Inverter (3Ph) — REeFI	5 kVA – 300 kVA	3-Phase
Batteryless Solar Hybrid PCU	5 kVA – 300 kVA	1Ph & 3Ph
HF (Transformerless) Solar Hybrid Inverter	3 kW–4 kW/24VDC & 5 kW–10 kW/48VDC	Wall-mounted, 1Ph
On-Grid Solar Inverter	2 kW – 6 kW (1Ph) / 5 kW – 25 kW (3Ph)	No storage
Online UPS (1Ph-1Ph)	1 kVA – 10 kVA	ETXi series
Online UPS (3Ph-1Ph)	10 kVA – 80 kVA	ETXi series
Online UPS (3Ph-3Ph)	5 kVA – 600 kVA	ETXi/HTXi series
Industrial UPS (general)	1 kVA – 300 kVA	1Ph & 3Ph
Industrial Bidirectional Inverter (PCS)	50 kW – 1000 kW+ (modular 50kW blocks)	For BESS
Fuel Cell Inverter	5 kVA – 300 kVA	1Ph & 3Ph
Industrial Battery Charger (FCBC/SCR/IGBT)	24V/48V/110V/220V/360V up to 800A (some listings show up to 1000A)	Multiple charger types
Static Frequency Converter (50↔60 Hz)	1 kVA – 300 kVA	Bi-directional
Static Frequency Converter (50↔400 Hz)	1 kVA – 300 kVA	For marine/defence/aerospace/test labs
Servo Voltage Stabilizer — Air-cooled	1 kVA – 250 kVA (single & three phase ranges combined)	Indoor use

Product Category	Power Range (site-confirmed)	Notes
Servo Voltage Stabilizer — Oil-cooled	25 kVA – 1000 kVA	24/7 duty
BESS — Mini e-Storage	5 kWh – 20 kWh	Residential/small commercial
BESS — Hybrid Eco Series	10 kWh – 200 kWh	Commercial
BESS — All-in-One Hybrid e-Storage	50 kWh – 200 kWh	Compact cabinet
BESS — Plug & Play Storage Cabinet	100 kWh – 215 kWh	Modular
BESS — MWh Energy Storage Container	400 kWh – MWh scale	Utility/large C&I

This table should replace the shorter product list currently in your chatbot prompt — it is more accurate to what's live on the site today (e.g., Online UPS 1Ph starts at 1 kVA, not 5 kVA; Servo Stabilizer now goes as low as 1 kVA, not 30 kVA).

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3. PRODUCT ARCHITECTURE — DATA_PENDING FIELDS NOW FILLED

3.1 Online UPS (ETXi / HTXi Series, 5 kVA – 600 kVA, 3Ph-3Ph shown; 1Ph and 3-1 variants share the same core technology)

- **Input sources:** Utility grid (mains) via rectifier/PFC front end; optional DG-compatible input circuit.
- **Storage integration:** Lead-acid, VRLA, and Lithium-ion batteries (MODBUS-ready battery management).
- **Typical use case:** Mission-critical industrial/commercial loads — manufacturing, healthcare, data centers, IT infrastructure — wherever a momentary outage would cause equipment damage or data loss.
- **Key specs:** Input voltage 415V ±20% (3-Ph); output 415V ±1%; frequency 50 Hz ±0.1% (crystal-controlled); efficiency >75% standard, up to 92% with IGBT front end; input PF >0.99 with PFC; overload 125% for 60s / 150% for 5s; IGBT-based 32-bit DSP control; GSM/RS-485 monitoring.
- **Configurations available:** Standalone, Hot Standby (100% redundant, automatic load transfer), and N+1 Parallel Redundant (load sharing, scalable, automatic redundancy management).

- **Rectifier options:** SCR, PFC, IGBT front-end, or PFC+SCR — selectable based on power-quality requirements.

3.2 Industrial Bidirectional Inverter (PCS)

- **Input sources:** Grid (AC, for charging), solar, wind, diesel/gas generators, and battery (DC) — genuinely bidirectional.
- **Storage integration:** LFP, NMC, Lead-Acid, and Flow batteries; verified compatibility with battery brands including IPower, Tata Chemicals, Exide, BYD, CATL, LG Energy Solution, and Samsung SDI.
- **Typical use case:** The power-conversion "engine" of a BESS — peak shaving, time-of-use arbitrage, grid-services revenue, microgrid/black-start, renewable integration, and backup power, scalable from 50 kW to 1000 kW+ in modular 50 kW blocks.
- **Key specs:** DC voltage 400–1000V (configurable); AC 230V (1Ph) / 400–415V (3Ph); 50/60 Hz auto-adjustable; DC-AC efficiency up to 98%, AC-DC up to 97%, round-trip 95%+; response time <10ms; switching time <100ms (grid-connected to off-grid); overload 110% continuous / 150% for 10s; communication via RS485, Ethernet, 4G/Wi-Fi, MODBUS TCP/RTU; protection rating IP21 (IP54 optional outdoor).
- **Warranty:** 5 years standard, extendable to 10 years.

3.3 Industrial Battery Charger (FCBC / SCR / IGBT-based)

- **Storage integration:** Lithium-ion, Lead-Acid, and VRLA batteries.
- **Typical use case:** DC uninterruptible supply for industrial/utility DC loads (switchgear control, telecom, substations) at 24V/48V/110V/220V/360V, up to 800A (some listings show up to 1000A on larger models).
- **Key specs:** Transformer-based isolation; forced-air cooling with temperature-controlled fans; LCD/LED diagnostics display; enclosure IP20–IP65 (customizable); protections include overcurrent, overvoltage, short-circuit.
- **Variants:** FCBC (Float-Cum-Boost Charger), Dual FCBC, FC+FCBC combination, SCR-based, and IGBT-based chargers, plus a dedicated Lithium-Ion Battery Charger line.

3.4 Static Frequency Converter

- **Typical use case:** Precision frequency matching for imported/frequency-sensitive machinery — CNC and production lines, test/QC labs, medical imaging and lab equipment, data-center support infrastructure, and marine shore-power/defence/aerospace applications.
- **Key specs (50↔60 Hz range):** Frequency regulation ±0.5% (no load to full load); voltage regulation ±1.5%; input/output voltage 110VAC or 400VAC (customizable); power factor 0.7 lead to 0.7 lag; overload 120% of rated capacity; efficiency >95% (up to 97%); DSP-based IGBT technology; single- and three-phase models; built-in isolation transformer.

- **Also available:** A 50↔400 Hz variant for marine/defence/aerospace/industrial-testing use, same 1–300 kVA range.
- Static conversion advantages over rotary converters: >95% efficiency vs 75–85%, ±0.5% vs ±2–5% frequency regulation, no moving parts/minimal maintenance, silent operation, instant startup vs spin-up delay.

3.5 Servo Voltage Stabilizer

- **Input sources:** Utility grid; specifically engineered to also tolerate diesel-generator (DG) power (handles frequency variation, rapid load changes, and power-factor swings typical of DG sets).
- **Typical use case:** Protecting three-phase industrial equipment (CNC machines, pharma manufacturing lines, cold storage compressors, textile machinery, data-center HVAC, hospital equipment) from voltage fluctuations and imbalance.
- **Two build types:**
  - **Air-cooled:** 3 kVA – 250 kVA (some single-phase models start at 1 kVA), natural/forced air, ⩾97% efficiency, 10–15 year component life, best for indoor/moderate-duty (20–22 hrs/day) use.
  - **Oil-cooled:** 25 kVA – 1000 kVA, BS:335-standard transformer oil, ⩾98% efficiency, 15–20 year component life, designed for continuous 24/7 duty and higher ambient temperatures (up to 50 °C).
- **Balanced vs. Unbalanced configuration:** Balanced type uses one servo motor for all three phases (lower cost, needs balanced supply); Unbalanced type uses three independent servo motors, one per phase (higher cost, handles mixed/unbalanced loads and supply — recommended for most modern industrial sites).
- **Key specs:** Output 415V AC ±1% (adjustable ±6%), regulation accuracy ±0.5%, response time <15–20 ms, correction rate 40–70V/second, THD <2–3%.
- **Standard warranty:** 18 months, extendable up to 5 years.

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4. TECHNICAL SPECIFICATION LAYER — UNCHANGED FROM ORIGINAL PDF

(Retained as-is from the original knowledge document; not contradicted by the website.)

- Electrical comparative specs (Grid Frequency Range, Input PF, Output THDi, Transfer Time, Self-Consumption) for SunMagic+ REefi, EnerTron (GTI), and EnerCube Mini — see original PDF Section 5.
- Environmental specs (Operating Temperature, Acoustic Noise, Max Altitude, Relative Humidity) — see original PDF Section 5.
- Protection Features Database for SunMagic+ REefi, EnerTron GTI, EnerCube Mini, and E-Series — see original PDF Section 5.

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5. WARRANTY & SUPPORT TERMS (from live product pages)

Product Line	Standard Warranty	Extended Option
Bidirectional PCS / BESS	5 years	Up to 10 years
Servo Voltage Stabilizer	18 months	Up to 5 years
EnerTron Grid-Tie Inverter	8 years	—
E-Series HF Hybrid	Standard (unspecified)	8%/year extended warranty option

Response-time commitments (Bidirectional PCS / BESS support, per enertechups.com):

- Remote diagnostics: within 2 hours
- Onsite support, metro cities: 24 hours
- Onsite support, other locations: 48 hours
- Emergency response: prioritized

Support channels: 24/7 technical helpline, WhatsApp Customer Care & Support (+91 9373679255), email (support@enertechups.com), and online chat on the website. The FAQ layer of the original PDF cites a slightly broader SLA (48–72 hours for standard toll-free service calls) — both figures can be kept, since they refer to different product tiers (BESS/PCS gets the faster 24–48 hr SLA; general toll-free service calls quote 48–72 hrs).

Service network: enertechups.com references 20+ / 100+ service centers depending on the page (BESS page says "20+ Service Centers Across India"; Servo Stabilizer page says "100+ service centers pan-India"). Recommend confirming the current, single accurate count with the service team before publishing externally.

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6. INSTALLATION KNOWLEDGE LAYER — UNCHANGED FROM ORIGINAL PDF

(Pre-installation checklist, site requirements, mechanical/electrical installation, earthing structure, commissioning steps — see original PDF Section 7. No conflicting data found on the website.)

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7. USER MANUAL LAYER — UNCHANGED FROM ORIGINAL PDF

(SunMagic+ REefi operating modes 1–4, control interface/display parameters, remote override via Modbus RTU — see original PDF Section 8.)

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8. WIRING LOGIC LAYER — UNCHANGED FROM ORIGINAL PDF

(AC-coupled vs DC-coupled BESS topologies, grid wiring, DC circuit safety — see original PDF Section 9.)

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9. PRODUCT COMPARISON ENGINE — EXTENDED

(Original four entries retained; two more added from website content.)

- **Application Requirement:** Frequency-mismatched imported machinery (50Hz vs 60Hz origin) **Recommended System:** Static Frequency Converter **Reason/Differentiator:** Bi-directional 50↔60 Hz conversion, >95% efficiency, no moving parts, ±0.5% frequency regulation — avoids re-engineering imported equipment.
- **Application Requirement:** Mission-critical 3-phase industrial UPS load with future capacity growth **Recommended System:** Online UPS (ETXi/HTXi, Hot Standby or N+1 parallel configuration) **Reason/Differentiator:** True online double-conversion with zero transfer time; N+1 parallel redundancy allows scaling capacity without replacing the whole system.

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10. FAQ LAYER — EXTENDED WITH WEBSITE FAQs

Pre-Sales & Company

- Q: How long has EnerTech been in business? A: The company states 35+ years of operation (see Section 1.1 for the 1989/1999 discrepancy note).
- Q: Is EnerTech a trusted brand? A: Yes — 35,000+ installations across India and abroad, serving schools, hospitals, industries, and government projects.
- Q: Can EnerTech inverters run without batteries? A: Yes — the SunMagic+ REefi and E-Series HF hybrid inverters support battery-less operation (grid-tie style), with batteries addable later.

Servo Stabilizer

- Q: Balanced vs unbalanced servo stabilizer — which do I need? A: Balanced type (single motor) suits symmetrical loads with balanced supply; unbalanced type (three independent motors) suits mixed or unbalanced loads and is recommended for most modern industrial sites.
- Q: Do servo stabilizers work with diesel generators? A: Yes, they include electronic circuitry specifically for DG compatibility (frequency variation, rapid load change tolerance).

Bidirectional PCS / BESS

- Q: What's the difference between a PCS and a regular inverter? A: A regular inverter only converts DC→AC; a bidirectional PCS converts both ways and adds battery management, grid sync, and energy optimization.
- Q: Typical installation timeline? A: 4–6 weeks from order to commissioning, including permits, delivery, installation and testing.

Static Frequency Converter

- Q: Can the converter be customized? A: Yes — input/output frequency range, power rating, and control features can be tailored per application.

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11. INDICATIVE PRICING (Website "Sale" Listings — for internal sales reference only, NOT to be quoted directly to customers without confirmation)

These are illustrative list prices pulled from the live shop pages on 25 Jul 2026. Prices change frequently — always confirm current pricing with the sales team (9370659050 / sales@enertechups.com) before quoting a customer.

Product	List Price	Sale Price
HF Solar Hybrid PCU 3kW/24VDC 1Ph	₹85,088	₹56,725
HF Solar Hybrid PCU 10kW/48VDC 1Ph	₹2,45,339	₹1,63,559
On-Grid Inverter 3kW 1Ph	₹32,175	₹21,450
On-Grid Inverter 6kW 1Ph	₹54,210	₹36,110
Mini e-Storage 3kW/5kWh	₹3,48,550	₹2,32,700
Mini e-Storage 10kW/20kWh	₹7,50,000	₹6,10,000
Online UPS 20kVA 3-3Ph	₹3,63,720	₹2,42,480
Online UPS 60kVA 3-3Ph	₹7,43,235	₹4,95,490
Static Frequency Converter 10kVA (50-400Hz)	₹3,71,545	₹2,47,696
Static Frequency Converter 50kVA (50-400Hz)	₹11,68,736	₹7,79,157
Static Frequency Converter 10kVA (50-60Hz)	₹3,70,000	₹1,67,700
Static Frequency Converter 60kVA (50-60Hz)	₹7,50,000	₹5,30,000

Bidirectional PCS / BESS indicative project economics (typical 250kW PCS + 500kWh battery commercial installation):

- Total investment: ₹2.15–2.95 crore (PCS ₹50–75L, battery ₹1.5–2Cr, install/engineering ₹15–20L)
- Annual benefit: ₹45–65 lakh/year (peak-demand reduction + TOU arbitrage + grid-services revenue)
- Typical ROI: 3–5 years; 10-year lifecycle net benefit ₹1.55–3.55 crore; IRR 20–35%

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12. APPLICATIONS BY INDUSTRY (from Servo Stabilizer & Bidirectional PCS pages — applies broadly across the power-product range)

- **Heavy manufacturing/metal:** CNC machines, injection molding, rolling mills, welding, laser cutting, industrial robots.
- **Textiles:** Spinning, weaving looms, knitting, dyeing, printing, embroidery.
- **Pharmaceutical/healthcare:** API manufacturing, tablet presses, capsule filling, autoclaves, clean-room HVAC, CT/MRI/X-ray, lab analyzers.
- **Food & beverage:** Cold storage, dairy processing, flour/oil mills, bottling, meat processing, bakery equipment.
- **Printing & packaging:** Offset/flexo/gravure/digital presses, lamination, die-cutting, carton making.
- **Petrochemical/chemical:** Reactors, distillation columns, compressors, mixers, heat exchangers.
- **Paper & pulp, rubber & plastics, automotive component manufacturing.**
- **Commercial/infrastructure:** Malls, hotels, hospitals, data centers, corporate offices, airports, metro stations, telecom towers.
- **Agriculture:** Rice/dal mills, poultry farms, cold chain, greenhouses, irrigation pumps.
- **BESS-specific:** Manufacturing facilities, office/corporate campuses, data centers, hotels/hospitality, renewable (solar/wind) integration, microgrids, utility-scale grid stabilization, T&D deferral, residential/community storage, housing societies.

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13. CUSTOMER SUPPORT BEHAVIOR LAYER — UNCHANGED FROM ORIGINAL PDF

(Tone guidelines, clarifying-question patterns, escalation triggers for over-temperature/smoke-detection alarms and repeated grid-sync failure — see original PDF Section 13.)

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14. SUPPORT PROCESS LAYER — UNCHANGED, WITH ADDED DETAIL

(Pune Head Office remote support team, video-call resolution — see original PDF Section 14. Section 5 above adds the specific response-time SLAs now published on the website.)

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15. ITEMS TO CONFIRM WITH THE CLIENT BEFORE FINALIZING THIS AS YOUR RAG SOURCE

1. **Founding year:** 1989 vs 1999 (site inconsistency — see 1.1).
2. **Total installed capacity:** 150 MW (original PDF) vs 600+ MW (current homepage) — likely the site was updated; confirm which figure to use going forward.
3. **Number of service centers:** "20+" (BESS page) vs "100+" (Servo Stabilizer page).
4. **Sales contact number:** 9370659050 (used in your chatbot prompt) does not appear on the public site (public sales lines are 9373336340 / 9822407189) — confirm it's a valid, currently-monitored number for Mr. Shivaji Chouhan.
5. **Pricing table in Section 11** is a snapshot only and will go stale quickly — if you want live pricing in the chatbot's knowledge base, it should be refreshed on a schedule (e.g., weekly) rather than hard-coded.